Lexical Functions to Improve Lexical Competence

Na An[1], Gabriel Beaudoin[2], Noah Tremblay Taillon[3]
[1] na.an1@umontreal.ca, [2] gabriel.beaudoin.3@umontreal.ca,
[3] noah.tremblay.taillon@umontreal.ca

Abstract

Large language models exhibit strong performance on many natural language processing tasks, yet their ability to model fine-grained lexical relations remains limited compared to that of human native speakers. Recent work has used the Meaning-Text Theory (MTT) and lexical functions to evaluate models' lexical competence, but has focused primarily on evaluation rather than on improvement. In this project, we construct a structured dataset of French lexical units annotated with lexical functions derived from the French Lexical Network, and use it to fine-tune and evaluate three language models. We investigate whether explicit training on a structured lexical representation improves models' lexical competence and generalization beyond memorized lexeme pairs. Our results show that structural injection improves lexical competence for Qwen2.5-7B (up to 3×), but effectiveness varies; Mistral-7B exhibits catastrophic forgetting and FlauBERT shows minimal gains. No consistent improvements are observed on the COLE benchmark, suggesting that improvements in structured lexical knowledge do not readily transfer to broader NLU capabilites.

1 Introduction

Modern natural language processing systems, particularly Large Language Models (LLMs), achieve strong performance across a wide range of tasks. However, their ability to model fine-grained lexical relations-specifically, word meaning and semantic relations between lexical units-remains limited. To address this limitation, we draw on the Meaning-Text Theory (MTT) (Mel'cuk, 1997), which formalizes systematic relations between lexical units through Lexical Functions (LFs). This allows the MMT to create a graph of a language, where the nodes are lexical units and the arcs are Lexical Functions. An example of LF would be *Magn*, which intensifies the input; *Magn(rain) = heavy*.

While previous research has primarily utilized LFs to evaluate models' lexical competence, we investigate whether explicitly injecting LF-based structure into training can fundamentally improve LLMs' natural language understanding (NLU). We construct a structured dataset derived from the French Lexical Network (fr-LN) and fine-tuned three model architectures. We compare structured lexical supervision with standard text-based training to assess its impact on lexical competence and generalization.

In summary, our experiments reveal that while fine-tuned models can achieve a higher understanding of structured lexical relations, notably for Qwen2.5-7B, these gains are architecture dependent. We observe a trade-off between specialized lexical competence and general stability, as evidenced by the catastrophic forgetting observed in Mistral-7B and the minimal improvement of FlauBERT.

2 Related Work

To evaluate language models' NLU, benchmarks must be sufficiently general to to capture diverse aspects of models' linguistic capabilities (Ni et al., 2025). GLUE (Ni et al., 2025) introduced the first unified benchmark, providing a single aggregated score to quantify model performance across tasks. Building on this idea, FLUE, a French specific benchmark composed of six linguistic evaluation tasks, was proposed (Le et al., 2020). To further increase diversity and enable more fine-grained evaluation, COLE was introduced as a comprehensive benchmark with 23 tasks and a unified aggregated score (Beauchemin et al., 2025).

While general NLU benchmarks are useful, they do not support fine-grained evaluation of specific linguistic capabilities, such as lexical competence.

The French Lexical Network (fr-LN) (ATILF, 2025) is a formal French lexicon grounded in the

MTT, describing in total 29, 976 lexical units. LFs encode relations between these units, forming a graph-structured representation. Using this resource, Liu and Lareau (2024) proposed an evaluation framework to assess CamemBERT's (Martin et al., 2020) sensitivity to French idioms. Petrov et al. (2025) proposed the Q&A-LF benchmark to evaluate models' lexical knowledge using LFs. Using template-based prompts for 15 selected lexical functions, the authors show that models generally perform well on tasks relying on morphological cues, but struggle with those involving world conception and compositional reasoning, demonstrating the lack of structured semantic mapping. Likewise, Li (2025) evaluate models using a broader set of 82 LFs.

Existing MTT-based evaluation approaches tend to exclude rare LFs in their testing, in order to have a more stable and general score. To our knowledge, no prior work has explored using LF-based representations to explicitly enhance model performance.

3 Method

We adopt a parameter-efficient fine-tuning approach to verify if explicit structural injection of LFs improves linguistic competence. Our primary objective was to compare the performance of fine-tuned models against their base counterparts to determine if explicit training yields a better understanding of abstract lexical relations.

To achieve this, we rely on Low-Rank Adaptation (LoRA) (Hu et al., 2021), which enables efficient adaptation by updating a small number of parameters, allowing us to inject specific MTT-based constraints without full model retraining.

We apply this methodology to three distinct architectures: Mistral-7B (Jiang et al., 2023), Qwen2.5-7B (Yang et al., 2024), and FlauBERT (Le et al., 2020). These models are selected to cover complementary design axes, including multilingual instruction-tuned LLMs and a French-specific encoder model. This diversity, while keeping model size manageable for academic experimentation, enables us to assess how LF-based supervision affects models with different architectures and scales.

4 Experiments

The code and datasets for this project are publicly available at `https://github.com/GabrielBeaudoinUdem/ift6289-project`. The repository includes instructions for environment setup and experiment reproduction.

4.1 Datasets

We construct a structured dataset based on the French Lexical Network (fr-LN) (ATILF, 2025), formatted in JSON. Each entry includes its part of speech (POS), sense identified to account for polysemy, associated LFs, an example illustrating its usage in context and its group ID. Although all of the data are available in fr-LN, our contribution lies in restructuring it appropriately into a format suitable for model training, rather than using it purely for evaluation. To ensure sufficient coverage and consistency across the dataset, we apply a minimum frequency threshold on LFs (since some LFs have very few instances in fr-LN). The data set is divided into 80%/20% train-test sets for our experiments.

4.2 Baselines

Our primary baselines consists of the original, unfinetuned versions of the three selected models (Mistral-7B, Qwen2.5-7B, and FlauBERT). We evaluate the base models in a zero-short setting, i.e., without any task-specific fine-tuning or weight updates.

Additionally, to ensure that any observed performance gains stemmed specifically from the injection of structured lexical knowledge rather than simply from further exposure to the French language, we introduced a secondary baseline by fine-tuning each model on a comparable volume of unstructured, raw French text extracted from Wikipedia. Comparing LF-finetuned models with the Wikipedia-finetuned baseline allows us to isolate the impact of the structured MTT-based supervision from that of additional French text exposure.

4.3 Evaluation Methods

We perform a two-step evaluation. First, we conduct a supervised-learning-inspired evaluation on the 20% test split. Following the evaluation procedure of Petrov et al. (2025), we design fixed query templates for every LF. For example, a prompt for the S_0 function is *"What noun corresponds to the verb «Verb»?"*, expecting a single lexical unit as output.

We evaluate model outputs using Exact Match (EM) and Contain Match (CM) (Petrov et al., 2025), along with Accuracy and F1 score. CM captures partial correctness when the predict output contains

the target, which is particularly useful given that models would violate the single lexical unit constraint. As the second step, we evaluate models on the COLE benchmark (Beauchemin et al., 2025) to assess whether learning LFs enables better general understanding of natural language beyond specific LF tasks.

4.4 Experimental Results

We adopt LoRA instead of full fine-tuning, as it enables efficient injection of MTT constraints without updating all model parameters. We implement all experiments using the HuggingFace Transformers and PEFT libraries. For Mistral-7B-v0.1 and Qwen2.5-7B, we set the LoRA with rank $r = 16$ and scaling factor $\alpha = 32$, applied the query, key, value, and output projection layers. Both models are trained for 1 epoch on local MPS hardware with a learning rate of 1×10^{-4} and an effective batch size of 16, using a subset of 5,000 examples and their respective instruction templates (`[INST]` and ChatML). FlauBERT was fine-tuned for 3 epochs on CUDA hardware with a learning rate of 3×10^{-5} using a template based on MLM. All models are trained with the AdamW optimizer and gradient checkpointing to manage memory constraints.

4.5 Results and Analysis

	FlauBERT Base	FlauBERT Wiki	FlauBERT LF	Qwen2.5 Base	Qwen2.5 Wiki	Qwen2.5 LF	Mistral Base	Mistral Wiki	Mistral LF
A_0	2.7	8.2	3.9	12.9	6.2	67.2	58.6	60.9	25.0
A_1	5.6	12.5	9.7	16.9	16.1	34.7	9.7	7.7	7.3
A_2	9.1	9.1	9.1	6.8	6.8	34.1	18.2	31.8	13.6
Able_1	4.5	15.9	6.8	0.0	0.0	38.6	13.6	4.5	11.4
Able_2	6.2	6.2	7.8	4.7	1.6	57.8	37.5	40.6	18.8
Adv_0	2.6	2.2	2.6	12.1	2.2	76.3	38.8	56.0	22.4
Adv_1	25.0	14.3	10.7	10.7	0.0	60.7	46.4	46.4	14.3
Anti	4.4	10.3	10.3	5.9	4.4	58.8	34.6	47.8	24.3
AntiBon	11.1	13.9	12.5	0.0	2.8	12.5	11.1	11.1	0.0
AntiMagn	14.3	21.4	17.9	0.0	0.0	0.0	3.6	7.1	3.6
AntiVer	16.7	16.7	16.7	0.0	0.0	4.2	0.0	4.2	4.2
Bon	35.7	46.4	25.0	0.0	0.0	0.0	0.0	0.0	0.0
Contr	3.4	3.4	3.4	3.4	2.2	28.0	30.2	34.1	10.3
Conv_21	11.1	30.6	33.3	13.9	12.5	11.1	11.1	8.3	6.9
Fact_0	12.5	12.5	12.5	0.0	0.0	21.9	9.4	12.5	15.6
Fact_2	14.1	18.8	21.9	0.0	0.0	53.1	14.1	12.5	21.9
Gener	2.1	5.6	5.6	25.7	36.1	32.6	31.9	32.6	24.3
Hypo	10.0	32.5	10.0	10.0	5.0	7.5	2.5	2.5	5.0
Loc_in	50.0	50.0	50.0	20.8	29.2	54.2	87.5	79.2	37.5
Magn	3.3	7.6	8.7	3.3	2.2	4.3	6.5	5.4	1.1
Mero	8.3	8.3	14.6	6.2	4.2	18.8	29.2	31.2	8.3
Mult	9.1	9.1	9.1	9.1	4.5	15.9	31.8	29.5	22.7
Oper_1	4.3	3.4	3.4	0.4	1.3	15.9	8.2	8.6	4.7
Real_1	5.7	4.7	6.2	1.6	0.5	24.5	14.6	16.7	11.5
Redun	7.8	12.5	9.4	0.0	3.1	9.4	18.8	10.9	0.0
S_0	3.0	3.4	4.1	24.3	15.5	39.2	43.9	41.9	12.8
S_1	3.8	5.3	4.9	11.7	9.5	20.5	26.5	25.8	13.6
S_2	3.2	3.2	3.2	8.9	10.5	19.4	15.7	16.5	6.0
S_3	7.1	7.1	7.1	15.2	5.4	13.4	15.2	18.8	5.4
S_instr	13.9	13.9	19.4	22.2	11.1	22.2	33.3	30.6	22.2
S_loc	5.5	7.0	4.7	15.6	13.3	28.1	21.1	27.3	11.7
S_res	10.0	10.0	10.0	17.5	25.0	13.8	15.0	13.8	1.2
Sing	7.7	7.7	7.7	0.0	0.0	11.5	9.6	11.5	3.8
Syn	5.6	12.5	11.8	9.7	9.0	18.8	16.7	15.3	6.9
V_0	1.1	2.9	5.5	4.8	0.4	60.3	34.6	36.0	24.6
Ver	16.7	16.7	16.7	0.0	0.0	0.0	0.0	0.0	0.0
TOTAL (Moyenne)	5.7	8.2	7.6	9.9	7.7	33.4	25.4	27.3	12.9

Figure 1: Lexical Functions (LF) benchmark performance heatmap (percentage)

Evaluation on the Lexical Functions (LF) benchmark (see Figure 1) reveals clear differences in how model architetures respond to structural knowledge injection.

Qwen2.5-7B demonstrates the largest improvement under LF fine-tuning, with its overall score increasing from 9.9% (Base) to 33.4% (LF). The fine-tuned model performed exceptionally well on functions involving Part-of-Speech (POS) transformations, such as *S_0*, *V_0*, and *Adv_0*. However, this improvement appears to be driven in part by better alignment with task-specific output constraints, as the fine-tuned model more consistently produces single-lexeme answers required by the evaluation.

Conversely, Mistral-7B exhibits a severe performance drop after LF fine-tuning, decreasing from a robust 25.4% (Base) to 12.9% (LF). This indicates catastrophic forgetting and possible severe overfitting to the training distribution. Notably, Mistral Base and Mistral Wiki remain surprisingly competitive with the fine-tuned Qwen model. This competitiveness is largely driven by exceptionally high scores on a subset of functions (e.g., *Loc_in*), suggesting that global averages should be interpreted with caution.

FlauBERT remains relatively stable across all configurations, with scores remaining limited change (ranging from 5.7% to 8.2%). One possible explanation is that the learning rate is too low to induce meaningful parametric updates for this architecture, leading to a ceiling effect. Notably, FlauBERT Wiki achieved the highest score among the encoder variations. Its slight advantage on the *Hypo* (hyponym) function aligns with the encyclopedic nature of Wikipedia, where hypernym/hyponym relations are prevalent. However, this trend does not extend to other related functions, such as *Gener* function. This suggests that the observed gains are function-specific rather than indicative of a broader improvement in hierarchical lexical reasoning.

Finally, certain categories such as *Ver*, *AntiVer*, *AntiMagn*, and *Bon* remain consistently challenging across all models and configurations. This may reflect either limited representation of these relations in pre-training corpora or their instrinsic semantice complexity. Furthermore, the baseline comparison reveals that fine-tuning on unstructured Wikipedia data does not consistently outform the base models, confirming that simply increasing a model's exposure to French text is not sufficient to improve fine-grained lexical competence.

To evaluate whether explicit lexical training transfers to broader natural language understanding, we assess the models on the COLE benchmark (see Figure 2). We observe an obvious divergence between the encoder (FlauBERT) and the generative decoder architectures (Qwen2.5, Mistral).

FlauBERT achieves near-perfect scores (100%) on several classification tasks (e.g., *daccord*, *french_boolq*, *mms*, and *wino_x*), while Qwen and Mistral obtain 0.0% on these tasks. This discrepancy is likely an artifact of the evaluation setup, rather than evidence of a genuine capability gap between models. COLE tasks are primarily designed for encoder-based models that produce predefined labels. In contrast, generative LLMs must generate exact target token to be scored as correct. Without task-specific prompting, generative models often fail to produce the exact label required by the evaluation, resulting in zero scores despite potentially correct predictions. Consequently, COLE scores for the generative models likely underestimate their actual NLU capabilities.

Despite these formatting constraints, Qwen2.5-7B LF shows a modest increase in global score compared to its base version, driven by improvements on tasks such as *allocine* (sentiment analysis). FlauBERT remained mostly stable, though the LF exhibits an unexpected improvement on the *gqnli* (a natural language inference task distinguishing entailment, contradictions and neutral statements).

However, Mistral-7B's performance on COLE is consistent with the catastrophic forgetting observed in the LF evaluation. The Mistral LF-finetuned model obtains a global score of 3.50%, with 0.0% on most tasks, suggesting overfitting to the LF training distribution at the expense of its general language modeling ability.

Overall, these results point to a fundamental mismatch between evaluation formats: the COLE benchmark favors encoder-based models which are optimized for classification, while the LF query benchmark favors generative models designed to produce constrained lexical outputs. Neither benchmark alone provides a complete assessment of model capabilities. Our results indicate that while targeted LoRA fine-tuning can teach a generative

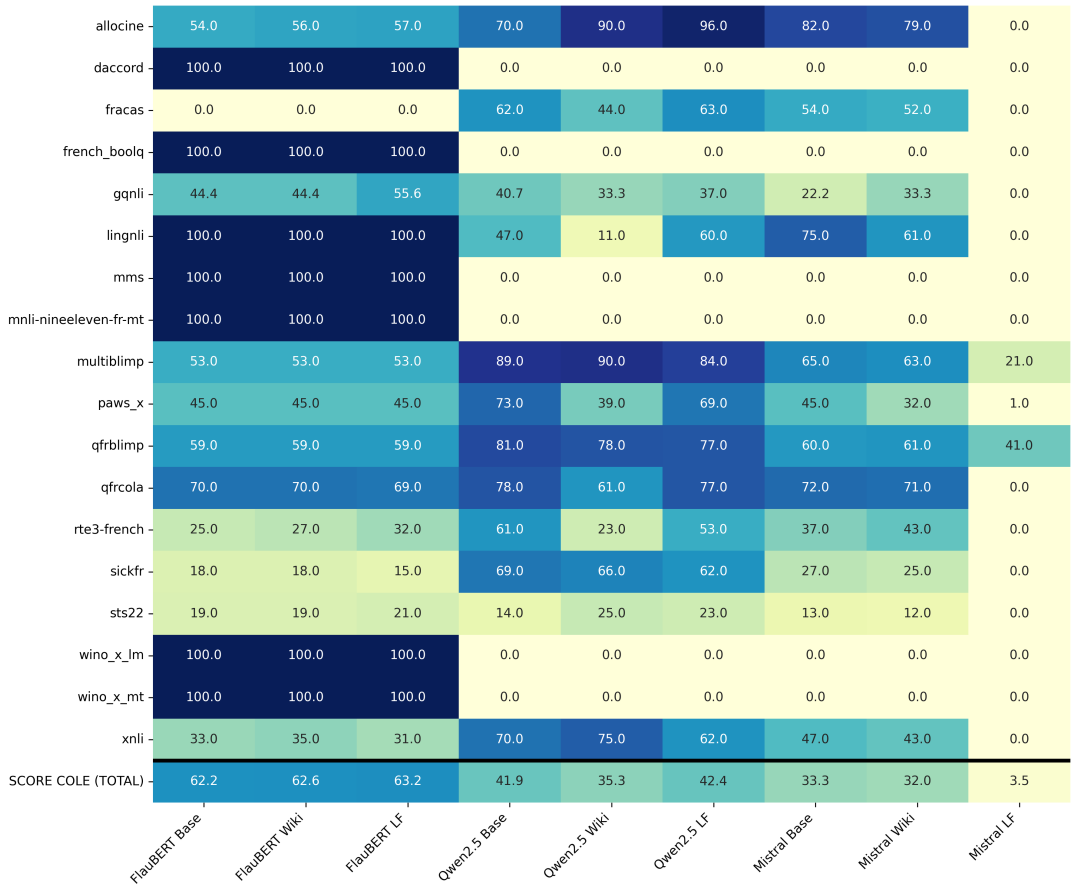

Figure 2: COLE benchmark performance heatmap (percentage)

model to excel at structural semantic mapping (as observed with Qwen), these gains do not readily generalize to broader NLU tasks and may come with a risk of catastrophic forgetting (as seen with Mistral).

5 Conclusion

In this project, we investigate whether explicitly training language models on structured lexical functions, derived from the Meaning-Text Theory, can improve both their specific lexical competence and their general Natural Language Understanding (NLU). To do so, we build a specialized LF dataset from the French Lexical Network and apply LoRA fine-tuning across three distinct architectures.

Our findings are mixed and highlight the complexity of augmenting LLMs with highly structured linguistic data. On one hand, we observe a resounding success with Qwen2.5-7B, whose performance on the LF benchmark more than triples. This suggests that structured lexical supervision can be effective when the model architecture and instruction-tuning paradigm are well aligned with the training objective.

On the other hand, we must remain highly cautious regarding the generalization of these results. Enhancing specific lexical knowledge did not translate into a broader improvement on the general NLU benchmark (COLE). Moreover, the severe performance degradation observed in Mistral-7B highlights the risk of catastrophic forgetting and overfitting to task-specific output formats when applying parameter-efficient fine-tuning on highly specialized datasets.

Future work should explore alternative integration strategies that mitigate these limitations. In particular, methods such as soft prompting, adaptive LoRA configurations, or incorporating LF supervision during pre-training or alignment stages (e.g., RLHF or DPO) may provide a more balanced trade-off between structured lexical knowledge and robust general language understanding.

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A Contributions of Each Team Member

- **Noah Tremblay Taillon:** Responsible for extracting, cleaning, and preparing the linguistic functions dataset. He also developed the evaluation pipeline and prepared the benchmarking infrastructure for the linguistic functions. Together with Gabriel Beaudoin, he contributed to the writing of the project proposal, the mid-progress report, the final report, and the poster.

- **Gabriel Beaudoin:** Responsible for preparing the Wikipedia baseline dataset and integrating the COLE benchmark. He also led the fine-tuning process and executed the evaluation scripts for the Mistral-7B and Qwen2.5-7B models. Together with Noah Tremblay Taillon, he contributed to the writing of the project proposal, the mid-progress report, the final report, and the poster.

- **Na An:** Responsible for the configuration and fine-tuning of the FlauBERT model, and conducted the performance evaluations for this architecture. She contributed to the writing of the final report.